



“Causality applies
only to a system
which is left
undisturbed.”

Paul Dirac

SPIN STATES

An electron can exist in a superposition of different states, such as spin up or spin down.



IN TWO PLACES AT ONCE

In classical physics, waves can be added together to form another wave (superposition). Similarly, quantum states—described by wave functions—can be combined to form another quantum state. This is known as quantum superposition. A quantum system that could be found in one of multiple states (e.g. an electron could have spin up or spin down, see p.66) can be described with a superposition of all these possible states.